travels along and then subsequently annihilates at (c), producing, once again, a single string. Finally, at (d), this string gives up its energy by dissociating into a pair of strings that head off in new directions. Because of the single loop in the center of Figure 12.4, physicists call this a “one-loop” process. As with the interaction depicted in Figure 12.3, a precise mathematical formula can be associated with this diagram to summarize the effect the virtual string pair has on the motion of the two original strings.

But that’s not the end of the story either, because quantum jitters can cause momentary virtual string eruptions to occur any number of times, producing a sequence of virtual string pairs. This gives rise to diagrams with more and more loops, as illustrated in Figure 12.5. Each of these diagrams provides a handy and simple way of depicting the physical processes involved: The incoming strings merge together, quantum jitters cause the resulting loop to split apart into a virtual string pair, these travel along and then annihilate one another by merging together into a single loop, which travels along and produces another virtual string pair, and on and on. As with the other diagrams, there is a corresponding mathematical formula for each of these processes that summarizes the effect on the motion of the original pair of strings.4

Moreover, just as the mechanic determined your final car-repair bill through a refinement of his original estimate of $900 by adding to it $50, $27, $10, and $.93, and just as we arrived at an ever more precise understanding of the motion of the earth through a refinement of the sun’s influence by including the smaller effects of the moon and other planets, string theorists have shown that we can understand the interaction between two strings by adding together the mathematical expressions for diagrams with no loops (no virtual string pairs), with one loop (one pair of virtual strings), with two loops (two pairs of virtual strings), and so forth, as illustrated in Figure 12.6.

An exact calculation requires that we add together the mathematical expressions associated with each of these diagrams, with an increasingly large number of loops. But, since there are an infinite number of such diagrams and the mathematical calculations associated with each get increasingly difficult as the number of loops grows, this is an impossible task. Instead, string theorists have cast these